# Array Division

Time limit: 1.00 s
Memory limit: 512 MB

CSES 1085 | Sorting and Searching | https://cses.fi/problemset/task/1085

You are given an array containing $n$ positive integers.

Your task is to divide the array into $k$ subarrays so that the maximum sum in a subarray is as small as possible.

## Input

The first input line contains two integers $n$ and $k$: the size of the array and the number of subarrays in the division.

The next line contains $n$ integers $x_1,x_2,\ldots,x_n$: the contents of the array.

## Output

Print one integer: the maximum sum in a subarray in the optimal division.

## Constraints

- $1 \le n \le 2 \cdot 10^5$
- $1 \le k \le n$
- $1 \le x_i \le 10^9$

## Example

| Input |
| --- |
| 5 3<br>2 4 7 3 5 |

| Output |
| --- |
| 8 |

Explanation: An optimal division is [2,4],[7],[3,5] where the sums of the subarrays are 6,7,8. The largest sum is the last sum 8.